

Fuji Manual

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Chapter 1

Concept

1.1 Configuration File

1.1.1 Definition

All files inside `config/fuji` directory are named `configuration file`

1.1.2 Types

Note: The types of configuration files

1. Control File A `control file` is used to control behaviours.

1.1. Main-Control File The **main-control file** refers to the `config/fuji/config.json` file, which is used to enable/disable a module.

1.2. Module-Control File Some modules will have their own control file, which is used to control the behaviour of the module.

2. User-Data File User-data files are used to store the data generated by the user.

1.2 Module

1.2.1 Definition

A module is used to provide a specific purpose.

Example: The purpose of modules

1. **ChatModule** provides chat-format customization.
2. **TpaModule** provides `/tpa` command.

1.2.2 Properties

The properties of a module are as follows:

1. **Can be disabled** You can disable a module completely in `main-control` file by setting the value of its `enable` key to `false`.
2. **Can work standalone** The code of a module is self-contained, there is no symbol reference to other modules.

1.2.3 Module Path

A module is identified by a unique module path.

Example: What a module-path looks like?

The module path of the module `tpa` is `tpa`.
The module path of the module `history` whose parent module is `chat`, is `chat.history`.
You will see a list of `enabled` modules identified by their `module path` at the server-startup process.

A module can have sub-module.

The relationship between `parent-module` and `sub-module` is relative, and there is nothing special about `sub-module`.

1.2.4 How to enable/disable a module?

You can enable/disable a module in `config/fuji/config.json` by setting the value of its `enable` key to `true/false`.

A module will be enabled if the following conditions are met:

1. The value of `common.debug.disable_all_modules` is `false`.
2. The required dependency mods are installed.
3. Its `parnet-module` is `enabled`.
4. The value of its `enable` key is `true`.

Example: How to enable a sub-module?

To make the module `chat.display` enabled, you need to enable `chat` module first.

Tip: How to know what modules are enabled?

Issue `/fuji inspect modules` command, to see what modules you have enabled.

1.3 Job

1.3.1 Definition

A job is some things will be done repeatedly.

1.3.2 Cron Expression as Trigger Rule

A language named cron language is used to define when a job should be triggered.

Tip: Don't write cron expression by hand. Use generator!

A cron expression looks like `0 * * ? * *`, means trigger the job every minute.
You can use the generator to generate a cron expression: <https://www.freeformatter.com/cron-expression-generator-quartz.html>

1.4 Regex

1.4.1 Definition

Regex is a language used to define the pattern of strings.

Some modules that use `regex`:

See [command_warmup](#)

See [command_rewrite](#)

See [command_cooldown](#)

See [teleport_warmup](#)

See [cleaner](#)

1.4.2 Reference

1. <https://regexr.com/>
2. <https://regex101.com/>

1.5 Placeholder

1.5.1 Definition

A `placeholder` is a string which will be replaced based on context.

Tip: What is the placeholder api in fabric platform?

There is a plugin named `PlaceholderAPI` in bukkit platform.
Also, there is a mod named `Text Placeholder API` in fabric platform.
They are different projects, but provides the same purpose.

1.5.2 Example

Example: Replace player name by context

The placeholder `%player:name%` will be replaced by the name of the contextual player.

1.5.3 Reference

1. <https://placeholders.pb4.eu/user/default-placeholders/>
2. <https://placeholders.pb4.eu/user/mod-placeholders/>

1.6 Identifier

1.6.1 Definition

An identifier is a string to name an object in minecraft.

1.6.2 Example

Example: The identifier of items

The identifier of apple is "minecraft:apple".
The identifier of diamond is "minecraft:diamond".

Example: The identifier of entities

The identifier of zombie is "minecraft:zombie".

1.6.3 How to query all identifiers of a type

Example: Query all identifiers of entity

```
/summon ...
```

Example: Query all identifiers of block

```
/setblock ...
```

Example: Query all identifiers of item

```
/give ...
```

Chapter 2

Command

2.1 Definition

A **command** is used in Minecraft to perform a specific action/function. Fuji provides lots of commands for various purpose. Commands are registered by a module. By default, all modules are disabled, so you have to enable modules in `config/fuji/config.json` file, and re-start your server to apply the changes.

2.2 What are the commands provided by this mod?

Tip: Query all commands registered by fuji

You may want to ask, what is the command list provided by fuji? The developers are lazy to write documents, so the document may outdated. However, you can issue the `/fuji inspect fuji-commands` command, to see all the commands registered by fuji.

Chapter 3

Permission

3.1 Definition

A **permission** is used to decide whether a **player** can do something or not.

3.2 Types

To make the discussion clearer, we define the types of **permission** as follows:

1. **level permission** A permission level is a non-negative number used in vanilla minecraft. The higher number means the higher authority.
2. **string permission** Usually, a **string permission** is introduced by a **permission plugin**, such as luckperms.

3.3 What is the permission system used by fuji?

3.3.1 Explanation

Fuji use the mojang's vanilla permission system, which is based on level permission.

As a convention, most of the commands registered by fuji, requires level permission to be 0 to use. Only a few of the commands require the level permission to be 4 to use.

Tip: Modify the default requirement of all commands registered by fuji

See [Configuration](#)

3.3.2 Set a string permission for a command

By default, fuji only use the level permission as the requirement of a command. However, if you want to use string permission for a command, you can use `command_permission` module, which is used to override the requirement of an existing command.

Example: Allow players to use /seed command

The command `/seed` provided by mojang requires level permission to be 3 to use. If you want to allow players to use `/seed` command, but you don't want to grant `op` for them. Then in this situation, you can grant the string permission for them: `/lp group default permission set fuji.permission.seed true`, which means that: set the requirement of command `/seed` to string permission `fuji.permission.seed`.

Example: Dis-allow players to use /list command

The command `/list` provided by mojang required level permission to be 0 to use. If you want to dis-allow players to use `/list` command, but because this command requires no string permission to use, so it's impossible to ban it via luckperms. In this situation, you can grant a string permission: `/lp group default permission set fuji.permission.list false` for them, which means that: set the requirement of command `/list` to string permission `fuji.permission.list`.

Example: Unset the override of requirement of the command

To undo the operation in the first example, just issue `/lp group default permission unset fuji.permission.seed`

3.4 Getting Started

Ensure you have enabled the `command_permission` module in `config/fuji/config.json` file, and re-start your server to apply the change.

Tip: Query the required permission for a command string

Let's say you want to grant the **proper permission** for `/back` command, so that your players can use it. Here comes the problem, what is the proper permission for this command? Okay, don't worry, just issue the `/command-permission describe back` command, and fuji will describe it.

Tip: Setup multiple permissions at one shot

A permission node may have children permissions. You may want to grant one permission and all of its children permission. Of course, the luckperms allows you to grant multiple permissions at one shot. See more in [command_permission](#).

3.5 Advanced Usage

3.5.1 The string permission is calculated with a context

The luckperms mod provides the **context** to calculate a string permission. A context is a predicate which accepts a player, and output a boolean value according to its conditions. Luckperms pre-define some useful context like:

1. **world context** : the player should in a specified world, so that this string permission is valid.
2. **temporal context** : current time should not exceed the expiration time, so that this string permission is valid.

3.6 Reference

1. https://minecraft.fandom.com/wiki/Permission_level

Chapter 4

Meta

4.1 Definition

A `meta` is a `key-value` pair.

Note:

Note that `meta` is introduced by `luckperms` mod, there is no `meta` in vanilla minecraft.

4.2 Example

Example: Set a meta for a group.

The `home` module supports the meta `fuji.home.home_limit`, which controls how many homes a player can create. To set the max homes limits to 3: `/lp group default meta set fuji.home.home_limit 3`

Example: Query all metas for a group.

```
/lp group default info
```

Chapter 5

Configuration

5.1 Main-Control File

5.1.1 Configuration

core The core options inside `config/fuji/config.json` will influence all modules.

debug

disable_all_modules Used to test the compatibility between fuji and other mods.

log_debug_messages Whether to log the debug level messages into the console. Use this only for debug purpose, or it will cause console spam.

backup Fuji will back up the `config/fuji` directory automatically before it loads any module.

max_slots How many backup should we keep?

skip The list of `path_resolver` to skip in backup.
Insert `modules/head` means skip the folder `config/fuji/modules/head`.

language

default_language The default language to use.

Tip: Enable multi-language support for fuji

See [language](#)

permission

all_commands_require_level_4_permission_to_use_by_default By default, most of commands registered by fuji requires level 0 permission to use. (e.g. `/afk`, `/home`, `/warp`). Some commands requires level 4 to use. (e.g. `/fly`, `/god`, `/more`). Enable this option will cause all commands registered by fuji require level 4 permission to use.

quartz Fuji use `quartz` library as scheduler, all the [Job](#) are managed by quartz. Quartz library use a language called `cron language` to define when to trigger a job.

logger_level The logger level for quartz. The logger level from high to low are: OFF, FATAL, ERROR, WARN, INFO, DEBUG, TRACE, ALL.

Example: Enable all logs for quartz

Set the value to **ALL** to display all the messages from quartz. It's recommended to set at least **WARN** level, to avoid **console spam**.

5.2 Module-Control File

You can read more about `module-control` file for each module in [Module](#)

Tip: Use `/fuji reload` command to hot reload the configuration files.

The `/fuji reload` command is provided by `fuji` module. See [fuji](#)

Chapter 6

Module

6.1 afk

6.1.1 Purpose

This module provides afk detection, afk event, afk name customization and afk effects.

6.1.2 How it works?

For each player, we have an `input counter` to track the last input time.

An input is as follows: mine blocks, movement.

The `afk checker` defined by cron will run and compare 2 consecutive value of the `input counter` associated with the player, if it's the same, then the player is considered as in afk.

In other words, if the `afk checker` is defined to run every 5 minutes, a player flagged as in afk is actually no input for 5--10 minutes.

6.1.3 Command

- `/afk`

6.1.4 Configuration

format The `tab list name` format when a player is afk

afk_checker

cron The cron to define how the `afk_checker` is triggered.

afk_event Execute commands on afk events.

Example: Kick a player if he enters afk state

```
"on_enter_afk": [  
  "send-broadcast <gold>Player %player:name% is now afk",  
  "kick %player:name% You are kicked because of afk."  
]
```

event Afk effects are applied if a player enters afk state.

invulnerable Immune to all damage?

targetable Can be targeted by a hostile entity?

moveable Can be moved if in afk state?

6.2 anti_build

6.2.1 Purpose

This module allows you to ban the interaction with some item/block/entity.

6.2.2 Anti Types

The types supported by this module are as follows:

1. break_block
2. place_block
3. interact_item
4. interact_block
5. interact_entity

Tip: Query identifiers

See [How to query all identifiers of a type](#)

6.2.3 Exapmle

Example: Ban TNT

```
add minecraft:tnt into place_block list
```

Example: Ban TNT but allow a specific player to use

```
/lp user <player> permission set fuji.anti_build.place_block.bypass.minecraft:tnt
```

Example: Ban to place any kind of block

```
add * into place_block list
```

6.3 back

6.3.1 Purpose

This module allows player to teleport back to **last teleport point** or **death point**.

6.3.2 Command

- `/back`

6.3.3 Configuration

ignore_distance If the player's teleportation destination is close enough, we ignore this teleportation.

6.3.4 Meta

1. `fuji.back.max_location_entries_to_save` The max back location entries to save.

6.4 chat

6.4.1 Purpose

This module provides chat related customization.

Note: The compatibility with `styled-chat` mod and other `chat-related` mods

All submodules except the `chat.style` module are designed to work with `styled chat` mod and other `chat-related` mods. You are free to use these modules with them!

6.4.2 Reference

1. [Text Placeholder API - default placeholders](#)
2. [Luckperms - prefix, suffix and meta](#)

6.4.3 Sub-Module

6.4.3.1 style

6.4.3.1.1 Purpose

Customize global chat style, and also allow players to use `/chat style` to set their per-player chat style.

6.4.3.1.2 Feature

1. You can use `mini-message language` to define complex format.
2. Besides the `server chat format`, each player can also set their `per-player chat format`.
3. This module doesn't **break** the vanilla chat events, so it can work with other chat related mods.
4. You can control the usage of color and event tags using permissions.

Tip: Write complex style using mini-language

You can use `mini-language` to write complex text.

See more:

1. <https://docs.advntr.dev/minimessage/format.html>
2. <https://placeholders.pb4.eu/user/quicktext>

6.4.3.1.3 Configuration

style The style for chat message.

6.4.3.1.4 Command

- `/chat style`

Example: Set prefix and suffix for players

Luckperms is required to set `prefix` and `suffix`.

After you installed `luckperms` mod, just issue `/lp group default meta setprefix <yellow>[awesome]` to assign prefix.

Don't forget to insert `%fuji:player_prefix%` and `%fuji:player_suffix%` in `sender` option in configuration file, and issue `/fuji reload`

6.4.3.1.5 Example

Example: Set per-player chat style

```
/chat style set prefix + %message% + suffix
```

Example: Reset per-player chat style

```
/chat style reset
```

Example: Allow players to use "<blue>" tag

```
/lp group default permission set fuji.style.chat.blue
```

Example: Allow players to use "" tag

```
/lp group default permission set fuji.style.chat.b
```

Example: Allow players to use all tags

All tags also including dangerous tags like "<click>" tag which can run commands on clicked!

```
/lp group default permission set fuji.style.chat.*
```

6.4.3.2 display

6.4.3.2.1 Purpose

This module allows players to share their main hand item, their inventory and their enderchest.

Tip: Works with Styled Chat mod

The `chat.display` module works with `Styled Chat` mod.

6.4.3.2.2 Purpose

- `/chat display item`
- `/chat display inv`
- `/chat display ender`

6.4.3.2.3 Placeholder

1. `%fuji:item%` display player's main-hand item.
2. `%fuji:inv%` display player's inventory.
3. `%fuji:ender%` display player's ender-chest.

6.4.3.2.4 Configuration

expiration_duration_s For each display data, how long should we save in the memory. Note that if a player shares its inventory items, then fuji will save a copy of his inventory data in the memory.

6.4.3.2.5 Example

Example: Define a chat shortcut to create display.

You can use `replace` module, to define chat shortcut like `"item"`, `"inv"` and `"ender"`, and replace them with display placeholders. In this case, you can type `"item"`, `"inv"` and `"ender"` in chat message direct, to create a display, without issuing the command.

6.4.3.3 mention

6.4.3.3.1 Purpose

Insert the player name in the chat message, and the target player will be mentioned and sound notified.

Note: Problems with Styled Chat mod

The `chat.mention` module partial work with `Styled Chat` mod. You can use `chat.mention` module with `Styled Chat` mod, the target player still get mentioned, but the `mention_format` will not be rendered.

6.4.3.3.2 Configuration

mention_player If you insert **Steve** in chat message, then the player named Steve will get audio mention.

sound The type of **sound** used to notify the mentioned player.

Tip: Query all identifiers of sound

```
/playsound ...
```

volume

pitch

repeat_count The sound repeat count.

interval_ms The interval between each repeat.

6.4.3.4 trigger

6.4.3.4.1 Purpose

Allow you to define magic spells in chat string, to execute commands.

Tip: Works with Styled Chat mod

The `chat.trigger` module works with `Styled Chat mod`.

6.4.3.4.2 Example

Example: Define a simple magic spell in chat string.

```
{
  "regex": "magic",
  "commands": [
    "say magic!"
  ]
},
```

Example: Define a complex magic spell with arguments in chat string.

```
{
  "regex": "i am (.+)",
  "commands": [
    "say You just said: $0",
    "say Hello $1"
  ]
},
```

Example: Define a shortcut for chat display module with `Styled Chat mod`

```
{
  "regex": "(?<=^|\\s)item(?:=\\s|$)",
  "commands": [
    "run as fake-op %player:name% chat display item"
  ]
},
```

6.4.3.5 replace

6.4.3.5.1 Purpose

Allow you to define `regex` to replace `chat` string with a given `chat` text.

Warning: Didn't work with Styled Chat mod

The `chat.replace` module does work with `chat.style` module. But if you are using `Styled Chat` mod to handle the chat style, then this module will not work. It's because the `Styled Chat` mod hacks the Minecraft internal chat processing flow, and making the `chat.replace` module failed to work. However, since `Styled Chat` mod does support you to define custom emotions, which provides the same purpose as `chat.replace` module. **So, if you are using `Styled Chat` mod, you didn't need the `chat.replace` module, you can disable it!**

6.4.3.5.2 Example

Example: Define a chat shortcut to evaluate a placeholder

```
{
  "regex": "(?<=^|\\s)uuid(?:=\\s|$)",
  "replacement": "<green>my uuid is %player:uuid%</green>"
}
```

6.4.3.6 spy**6.4.3.6.1** Purpose

Spy on specified chat types.

6.4.3.6.2 Command

- `/chat spy`

6.4.3.7 history

6.4.3.7.1 Purpose

Store `chat` messages as a history, and send them to a player joined the server.

Tip: Works with Styled Chat mod

The `chat.history` module works with `Styled Chat` mod.

6.4.3.7.2 Configuration

buffer_size How many chat messages should we save, so that we can send for a new-joined player.

6.5 cleaner

6.5.1 Purpose

This module provides the entity cleaner to remove specified entities automatically.

Note: Only use this module to clean some edge-case entity

Since the vanilla minecraft also has a cleaner to remove the item stack in the ground, so it's recommended to only use this module to clean some weak-loading entities, like: the sand item stack ...

6.5.2 Command

- `/cleaner clean`

Note: The cleaner will keep silent if cleans nothing

If the cleaner cleans nothing, then it will keep silent. (Which means you will not see any message in console, or in-game chat)

Tip: See what is cleaned in cleaner broadcast.

Hover your mouse on the cleaner broadcast, you can see what has been removed.

6.5.3 Configuration

cron The cron used to define the job to trigger `/cleaner clean`.

key2age The key is translatable key, which you can query in `en_us.json language file in minecraft 1.21`.

The translatable key of entity starts with `entity.minecraft`.

The translatable key of item starts with `item.minecraft` and `block.minecraft`.

The age is the existence time of the entity, the unit of age is game tick, which means $20 \text{ age} = 20 \text{ ticks} = 1 \text{ second}$.

The cleaner will only remove the entities whose translatable key equals key, and age greater equals the defined age, and the entity must not in the ignore list.

Example: Clean the sand-block entity lives longer than 60sec

```
"block.minecraft.sand": 1200
```

ignore Entities match the ignore list will not be cleaned.

ignore_item_entity Should we ignore all item entity.

ignore_living_entity Should we ignore all living entity?

If you want the cleaner to remove monster or animals, you should enable this option.

ignore_named_entity Should we ignore named entity. (With name tag, or name changed by anvil.)

ignore_entity_with_vehicle Like entity riding in some other entity, e.g. minecraft, pig or spider

ignore_entity_with_passengers

ignore_glowing_entity

ignore_leashed_entity

Note: The built-in safety rule

The cleaner will **always ignore** the following types:

1. player
2. any block attached entity (e.g. leash_knot)
3. any vehicle entity (e.g. minecart, boat ...)

6.6 color

6.6.1 Purpose

This module provides colorize for things.

6.6.2 Sub-Module

6.6.2.1 sign

6.6.2.1.1 Purpose

Colorize the sign.

6.6.2.1.2 Configuration

requires_corresponding_permission_to_use_style_tag By default, any player can use all style tags. Enable this options requires the player to has corresponding permission to use style tag. e.g. `"fuji.style.sign.<style-tag>"`

6.6.2.1.3 Example

Example: Write colorful text in sign

```
<red>This is the first line  
<rb>The second line  
<bold>The third line  
<i>The last line
```

6.6.2.2 anvil**6.6.2.2.1 Purpose**

Colorize the anvil.

6.6.2.2.2 Configuration

requires_corresponding_permission_to_use_style_tag By default, any player can use all style tags. Enable this options requires the player to has corresponding permission to use style tag. e.g. `"fuji.style.anvil.<style-tag>"`

6.7 command_advice

6.7.1 Purpose

This module allows you to create advice for existing commands, to decorate existing commands.

6.7.2 Example

Example: Execute commands before/after executing a target command

```
{
  "match_command_string_regex": "back",
  "only_valid_when_command_is_executed_by_player": true,
  "advice_type": "BEFORE_EXECUTING",
  "commands": [
    "run as fake-op %player:name% say before executing /back for
    ↪ %player:name%"
  ]
},
{
  "match_command_string_regex": "back",
  "only_valid_when_command_is_executed_by_player": true,
  "advice_type": "AFTER_EXECUTING",
  "commands": [
    "run as fake-op %player:name% say after executing /back for
    ↪ %player:name%"
  ]
},
}
```

Example: Decorate an existing command

```
{
  "match_command_string_regex": "heal",
  "only_valid_when_command_is_executed_by_player": true,
  "advice_type": "AFTER_EXECUTING",
  "commands": [
    "say spawn a fireworks for player %player:name%",
    "run as fake-op %player:name% summon firework_rocket ~ ~1 ~
    {FireworksItem:{id:firework_rocket,components:
    {fireworks:{explosions:[{shape:small_ball,colors:[I;15961002]}}}}}"
  ]
}
```

Example: Cancel the executing of a target command with success

```
{
  "match_command_string_regex": "say (.+)",
  "only_valid_when_command_is_executed_by_player": true,
  "advice_type": "CANCEL_WITH_SUCCESS",
  "commands": [
    "send-message %player:name% <rb>You just said:</rb> $1"
  ]
}
```

6.7.3 Reference

The function of this module is similar to [command_bundle](#)

6.8 command_alias

6.8.1 Purpose

This module allows you to define command alias, which redirect to the existing command node.

Note: A command node is identified by path

See also: <https://minecraft.fandom.com/wiki/Commands>

6.8.2 Example

Example: Shorten a existing command

The configuration create a command alias from `/r` to `/reply`

```
{
  "from": [
    "r"
  ],
  "to": [
    "reply"
  ]
}
```

Example: Shorten a existing command

The configuration create a command alias from `/sudo` to `/run as fake-op`

```
{
  "from": [
    "sudo"
  ],
  "to": [
    "run",
    "as",
    "fake-op"
  ]
}
```


6.8.3 What's more?

Tip: How can I define complex commands?

As you can see, the `command alias` module only support to **redirect** a simple command into another simple command. In other words, you can only use this module to create **alias** to a **existing command node**. Also, it's not allow to define **variable** inside a **command alias**.

If you want to define complex commands, use [`command\_bundle`](#) module.

6.9 command_attachment

6.9.1 Purpose

This module allows you to attach commands into itemstack.

6.9.2 Command

- `/command-attachment`

Note: The same item-stack shares the-same instance

If you hold stick * 64 in your main-hand, then all the sticks share the same attached commands.

6.9.3 Example

Example: Make a magic-stick which heals the player on clicked.

Hold a stick item in your main hand.

```
/command-attachment attach-item-one heal
```

Example: Make a magic-stick which gives one diamond on left clicked with use limit 3 and gives one gold_ingot on right clicked with use limit 5.

Hold a stick item in your main hand.

```
/command-attachment attach-item-one --maxUseTimes 3 --interactType LEFT give  
%player:name% minecraft:diamond 1  
/command-attachment attach-item-one --maxUseTimes 5 --interactType RIGHT  
give %player:name% minecraft:gold_ingot 1
```

Example: Make a magic-stick which gives one apple on clicked with use limit 3 without destroying the item.

Hold a stick item in your main hand.

```
/command-attachment attach-item-one --maxUseTimes 3 --destroyItem false give  
%player:name% minecraft:apple 1
```

Example: Query the attached commands in the mainhand item.

```
/command-attachment query-item
```

Example: Let an entity says hello on right click

```
/command-attachment attach-entity-one <entity-id> say hello %player:name%
```

Example: Say hello if you stepped on a specific block

```
/command-attachment attach-block-one 0 0 0 --interactType STEP_ON say hello  
%player:name%
```

Example: Make a portal on a specific block

```
/command-attachment attach-block-one 0 0 0 --interactType STEP_ON tpportals  
--targetPlayer %player:name% --dimension minecraft:the_end --x 0 --y 66 --z  
0 %player:name%
```

6.10 `command_bundle`

6.10.1 Purpose

This module allows you to create bundle commands, input one command, output many commands.

6.10.2 Command

- `/command-bundle`
The root command of this module.

6.10.3 Feature

1. a user-friendly DSL to create new custom commands easily, with the interoperation of `variables`, `placeholders` and `selectors`.
2. support complex `command argument type`: `required argument`, `literal argument` and even the `optional argument` with a default value.
3. a powerful type-system to ensure the `type-safe` input, with fully command suggestion.

Tip: To query all type strings

```
/fuji inspect argument-types
```

4. register and un-register commands on the fly, without the server restart!

Example: Reload the bundle commands

Each time you modify the configuration file, you should issue `/fuji reload`, this will unregister all bundle commands in the server, and register the bundle commands defined in the file into the server.

Also, you can use the `/command-bundle un-register` and `/command-bundle register` manually.

6.10.4 Example

Example: Combine commands into one command

```
{
  "requirement": {
    "level": 0,
    "string": null
  },
  "pattern": "composite-heal",
  "bundle": [
    "say before heal %player:name%",
    "run as fake-op %player:name% particle minecraft:heart ~ ~2 ~",
    "run as player %player:name% heal",
    "say after heal %player:name%"
  ]
}
```

Example: Transform a command

```
{
  "requirement": {
    "level": 4,
    "string": null
  },
  "pattern": "warn <player player-arg> <greedy greedy-arg>",
  "bundle": [
    "run as player %player:name% send-message $player-arg <red>You are",
    ↪ warned: $greedy-arg"
  ]
}
```

Tip: Assign a string permission for a bundle command

See [Permission](#)

6.10.5 Reference

1. <https://www.gamergeeks.net/apps/minecraft/particle-command-generator>
2. <https://learn.microsoft.com/en-us/minecraft/creator/documents/particleeffects?view=minecraft-bedrock-stable>

6.11 command_cooldown

6.11.1 Purpose

This module provides:

1. **unnamed cooldown** : per command cooldown after the `command` execution
2. **named cooldown** : support to associate a named cooldown with commands.

Tip: How to write regex language?

See: [Regex](#)

6.11.2 Command

- `/command-cooldown`

6.11.3 Placeholder

1. `%fuji:command_cooldown_left_time <named-cooldown>%` the left time for specified named cooldown
2. `%fuji:command_cooldown_left_usage <named-cooldown>%` the left usage for specified named cooldown

6.11.4 Example

Example: Create a named cooldown

```
/command-cooldown create example 3000
```

Example: Test a named cooldown

```
/command-cooldown test example <player> --onFailed "say false  
%fuji:command_cooldown_left_time 1%/%fuji:command_cooldown_left_usage 1%"  
say true
```

Example: Reset a named cooldown for a player

Note that this will only reset the `timestamp` associated with the player, the `usage` associated with the player will not be reset.

```
/command-cooldown reset example <player>
```

Example: Create a named cooldown with 3 max usage and 15 sec cooldown

```
/command-cooldown create example 15000 --maxUsage 3
```

Example: Create a named global cooldown for all players

A named global cooldown means that, all players shares the same cooldown, instead of per-player.

```
/command-cooldown create example 3000 --global true
```

Example: Create a non-persistent named cooldown

A non-persistent named cooldown means that, the `timestamp` associated with a player will not be persisted into the storage. That's to say, a server restart will forget all `timestamp`, but the `usage` associated with a player will always be persisted.

```
/command-cooldown create example 999999999999 --persistent false
```

Taken this example, it means that each time the server restarted, the cooldown will be available only once.

6.12 `command_debug`

6.12.1 Purpose

Execute a command and see debug info.

6.12.2 Command

- `/command-debug`

6.13 command_event

6.13.1 Purpose

Execute commands on specific events.

6.13.2 How it works?

When an event occurs, this module will execute commands as console with the contextual player. The contextual player will be used to parse placeholders.

6.13.3 Example

Example: Welcome the new-bie player

```
"on_player_first_joined": {  
  "command_list": [  
    "send-broadcast <light_purple>Welcome new player %player:name% to join  
    ↪ us!",  
    "kit give %player:name% <kit-name>",  
    "run as fake-op %player:name% rtp",  
    "delay 10 spawnpoint %player:name%"  
  ]  
},
```

6.14 `command_interactive`

6.14.1 Purpose

This module allows you to write commands in `sign block`.

6.14.2 How it works?

When a player `right click` a `sign block`, this module will check if the `sign block` contains the character `"/"`. If contains, then we `treat` as the player issue the command followed by the character.

6.14.3 Definition

A `sign block` that contains the character `"/"` is named an `interactive sign block`.

Note: How can i edit an interactive sign block?

You need to press `shift + right click` to edit an interactive sign.

6.14.4 Example

Example: Basic usage

```
/say hi %player:name%  
line 2 empty  
line 3 empty  
line 4 empty
```

Example: Add a prefix description text

```
prefix /say first  
/say the second  
/say hi %player:name%  
/say the last command
```

Example: Concat commands between lines

```
prefix /say this is  
the first /say and the  
second  
line 4 empty
```

6.15 command_meta

6.15.1 Purpose

This module provides commands to operate on commands.

6.15.2 Sub-Module

6.15.2.1 run

6.15.2.1.1 Purpose

This module provides `/run` command, which can run a command with context.

6.15.2.1.2 Command

- `/run`

6.15.2.1.3 Example

Example: Give random diamonds to online players

```
/run as console give @a minecraft:diamond %fuji:random 8 32%
```

Example: Give online players random diamonds

```
/run as console foreach give %fuji:escape player:name% minecraft:diamond  
%fuji:escape fuji:random 8 32 1%
```

Example: Execute a command as a player

```
/run as player <player> back
```

Example: Execute a command as fake-op

```
/run as fake-op <player> give %player:name% minecraft:apple 1
```

6.15.2.2 for_each

6.15.2.2.1 Purpose

This module provides /foreach command. If a command is only targeted for single player, you can use /foreach to apply it for each player online.

6.15.2.2.2 Command

- /foreach

6.15.2.2.3 Example

Example: Say hello to online players

```
/foreach say hello %player:name%
```

Tip: Escape the placeholder properly

If you use foreach in scheduler module, then you should escape (Write %fuji:escape player:name% instead of %player:name%) the placeholder.

It's because the command-scheduler module will try to parse the placeholder, and you need to escape the placeholder, so that the placeholder can be parsed by /foreach command.

Here is an example about escape the foreach command in scheduler command list: /foreach give %fuji:escape player:name% minecraft:diamond 16

6.15.2.3 one_of**6.15.2.3.1 Purpose**

The one-of command will pick a random command from commands, and execute it as console.

6.15.2.3.2 Command

- `/one-of`

6.15.2.3.3 Example

Example: Pick a lucky command.

```
/one-of say 1 one-of say 2 one-of say lucky %player:name%
```

6.15.2.4 chain

6.15.2.4.1 Purpose

A chain command allows you to run another 2 commands, the first is any command, and the second is the chain command.

Note: The return value of a command

In vanilla minecraft, the return value of command, is an integer:

1. <0 failed
2. $=0$ passed
3. >0 success

6.15.2.4.2 Command

- `/chain`

6.15.2.4.3 Example

Example: A nested chain.

```
/chain say 1 chain say 2 chain say 3
```

Example: A breakable chain.

```
/chain bad command here chain say 2
```

Example: Use chain command with predicate command

```
/run as player <player> chain test-level-perm %player:name% 4 chain say  
value is true
```

6.15.2.5 `delay`

6.15.2.5.1 Purpose

Delay command allows you to execute a command in specified seconds.

Note: Delay command is only used for short term job

The command `/delay` is only used to perform short-term job, and will not be persisted on server shutdown. If you want to define long-term job, using `command_scheduler`

6.15.2.5.2 Command

- `/delay`

6.15.2.5.3 Example

Example: A basic usage

```
/delay 3 say three seconds passed
```

Example: A nested delay

```
/delay 1 delay 2 delay 3 say 6 seconds passed.
```

6.15.2.6 json

6.15.2.6.1 Purpose

Provides a unified json editor.

6.15.2.6.2 Command

- `/json`

6.15.2.6.3 Exaxmple

Example: Read a key

```
/json read "config/fuji/config.json" "$.core.quartz.logger_level"
```

Example: List keys

```
/json read "config/fuji/config.json" "$.modules.keys()"
```

Example: Set a key

```
/json write "config/fuji/config.json" "$.core.quartz.logger_level" STRING  
INFO"
```

6.15.2.6.4 Reference

1. <https://goessner.net/articles/JsonPath/>

6.15.2.7 attachment**6.15.2.7.1 Purpose**

Provides a unified attachment facility, which can attach any data to any object.

6.15.2.7.2 Command

- `/attachment`

6.15.2.7.3 Example

Example: Set a attachment

```
/attachment set news today hello world
```

Example: Get a attachment

```
/attachment get news today
```

6.15.2.8 shell

6.15.2.8.1 Purpose

This module provides `/shell` command, which executes the command line in the host shell.

Danger: This is a dangerous module

This module is a powerful and dangerous module, not recommended to enable it.

6.15.2.8.2 Configuration

enable_warning A precautionary option to prevent this module is enabled.

security The security options for this module.

only_allow_console Only allow the console to execute `/shell` command.

allowed_player_names Only allow the specified players to execute `/shell` command.

6.15.2.8.3 Example

Example: Create a file using placeholder

```
/shell touch %player:name%.dangerous
```

Example: Execute a program in the host os

```
/shell emacs
```

Example: Backup the data of your server

You can use `shell` module with `command scheduler` module as a combo: define a job to execute the shell command in os to execute a program to backup the data of your server. See more: <https://rdiff-backup.net/>

Example: Possible to download a virus from Internet and execute it!

```
/shell ...
```

6.16 command_permission

6.16.1 Purpose

This module provides the customization of **the requirement of all commands**.

6.16.2 Command

- `/command-permission gui`
- `/command-permission verbose`
- `/command-permission describe`

6.16.3 How it works?

The vanilla minecraft use a command system named brigadier.

All the commands are registered, parsed and executed by brigadier.

In this system, all commands are build into a **tree structure**, that is to say, all commands are a direct or in-direct child of the **root command node**.

Example: What is the path of a specific command node?

For example, the command `/gamemode creative Steve` is composed by 3 command node:

1. `"gamemode"` = a literal whose name is `"gamemode"`
2. `"creative"` = an argument whose type is `gamemode`, its name is `"gamemode"` and its value is `"creative"`
3. `"Steve"` = an argument whose type is `player`, its name is `"target"`, and its value is `"Steve"`

We say that the command path of `/gamemode creative Steve`, is `["gamemode", "gamemode", "target"]`.

Tip: How to query the name of an argument

You can issue `/help gamemode` which will display the name of arguments. Or you can issue `/fuji inspect server-commands` to query the **command path** of all commands registered in the server.

Also, each **command node** has its **requirement**, which is a **predicate** to check if the **command source** can use the command node.

Tip: Query the command path of a command.

```
/lp group default permission set fuji.permission...
or /command-permission gui
or /fuji inspect server-commands
```

Tip: Use verbose mode to see what is happening

Issue `/command-permission verbose` to enable verbose mode, and then issue the **target command** in game, then see the logs in console. Now, you can issue `/lp group default permission set fuji.permission.<command path of target command>` to let `command_permission` module intercept and handle the requirement predicate of the target command.

Warning: Understand the parent permission node and its children node

It's recommend to modify the `apply-sponge-implicit-wildcard` option to `false` in `luckperms.conf`, to control the `permission` node more fine-grained.

Warning: Be careful with wildcard-permission and regex-permission

The `apply-regex` and `apply-wildcards` options in `luckperms.conf` file controls how `luckperms` calculates the permission state for a player. You have to understand what you are doing before you assign a `wildcard-permission` and `regex-permission` for a player.

In short, what the `command_permission` module does is simple, it only intercepts all command nodes, and checks whether the target player has the permission `fuji.permission.<command path of target command>`. Then it decides whether to allow or dis-allow the usage of this command based on the permission calculation result from `luckperms`, or just simply do nothing if no related positive/negative permission is assigned.

The complex permission calculation is done by `luckperms`, including `permission inheritance`, `permission wildcards` and `permission regex`.

6.16.4 Example

Example: Allow everyone to use `/gamemode` command

```
/lp group default permission set fuji.permission.gamemode true
```

Tip: Allow the client-side to use gamemode switcher menu

After you assign the `/gamemode` command permission for players, the client-side also requires to install a mod to bypass the client-side permission checking: <https://modrinth.com/mod/switcher>

Example: Allow everyone to use `/gamemode` command except the player Alice

```
/lp group default permission set fuji.permission.gamemode true
/lp user Alice permission set fuji.permission.gamemode false
```

Example: Only allow everyone to use `/gamemode spectator`

It's impossible to assign a single gamemode, since the command path of `/gamemode creative` and `/gamemode spectator` are both `"gamemode.gamemode"`.

Notice that the first `"gamemode"` in the command path, means the literal argument `"gamemode"`.

The second `"gamemode"` in the command path, means an argument, whose type is `gamemode`. This gamemode argument contains all the 4 gamemodes: adventure, creative, spectator and survival. That's the real reason why we can't assign a single gamemode for the command `/gamemode`.

If you really want to assign only 1 single gamemode for everyone, you can use `command_bundle` to create a new command, which only switch the gamemode of player into spectator.

Example: More examples

See [Permission](#)

6.17 command_rewrite

6.17.1 Purpose

This module allows you to use regex language to rewrite the command line a player issued.

6.18 command_scheduler

6.18.1 Purpose

Define jobs using cron expression to execute commands.

6.18.2 How it works?

A Job is defined by crons, with conditions like left times.

A job will be triggered if current time meets any of its cron expressions.

A commands will be chosen randomly from the commands list of the job and executed if the following conditions are met:

1. The enable of the job is true
2. The left times of the job ≥ 0

The commands will be executed as console.

6.18.3 Command

- `/command-scheduler trigger`
Immediately trigger a job.
- `/command-scheduler list`
List all defined jobs.

6.18.4 Example

Example: Define a job that executes commands 32 times on every minute

```
{
  "name": "example_job",
  "enable": true,
  "left_times": 32,
  "crons": [
    "0 * * ? * * *"
  ],
  "commands_list": [
    [
      "say 1 minute passed"
    ],
    [
      "say 60 seconds passed"
    ]
  ]
}
```

6.19 `command_spy`

6.19.1 Purpose

Log command execution event into the console.

6.19.2 Configuration

ignore Ignore the command that matches regex.

spy_on_console should we spy the command execution of the console?

6.20 command_toolbox

6.20.1 Purpose

This module provides some simple commands. (misc commands)

6.20.2 Sub-Module

6.20.2.1 bed

- `/bed`

6.20.2.2 extinguish

- `/extinguish`

6.20.2.3 feed

- `/feed`

6.20.2.4 fly

- `/fly`

6.20.2.5 god

- `/god`

6.20.2.6 hat

- `/hat`

6.20.2.7 sit

- `/sit`

6.20.2.8 heal

- `/heal`

6.20.2.9 lore

- `/lore`

Example: Set lore for item in mainhand

```
/lore set <rainbow>the first line<newline><bold><green>the second
```

6.20.2.10 more

- `/more`

6.20.2.11 ping

- `/ping`

6.20.2.12 realname

- `/realname`

6.20.2.13 nickname

- `/nickname`

6.20.2.14 repair

- `/repair`

6.20.2.15 reply

- `/reply`

6.20.2.15.1 How it works?

When a player use the command `/msg` or `/tell`, we track the usage, and allow the target player to use `/reply` to send message to the player recently msg he.

Tip: Create a command alias like `/r`

You can use `command_alias` module to create a command alias from `/r` to `/reply`.

6.20.2.16 seen

- `/seen`

6.20.2.17 suicide

- `/suicide`

6.20.2.18 top

- `/top`

6.20.2.19 trashcan

- `/trashcan`

6.20.2.20 tppos

- `/tppos`

6.20.2.20.1 Example

Example: Teleport to offline location of a player

```
/tppos offline <player>
```

Example: Specify another player as the target player

```
/tppos --targetPlayer <player>
```

Example: Use fixed teleport mode

When any of the following args is specified, the `/tppos` command will do a fixed teleport.

```
/tppos --x 32 --y 64 --z 128 --yaw 60 --pitch 90 --dimension  
minecraft:overworld
```

Example: Use random teleport mode

The `/tppos` command will do a random teleport if not doing a fixed teleport.

```
/tppos --centerX 0 --centerZ 0 --circle true --minRange 512 --maxRange 2048  
--minY 32 --maxY 128 --maxTryTimes 8 --dimension minecraft:overworld
```

6.20.2.21 warp

- `/warp`

6.20.2.21.1 Example

Example: Set a display name for a warp

```
/warp set-name <warp> <blue>This is the display name
```

Example: Set a lore for a warp

```
/warp set-lore <warp> <blue>This is the first line|<red>This is the second  
line
```

6.20.2.22 burn

- `/burn`

6.20.2.23 help-op

- `/help-op`

6.20.2.24 near

- `/near`

6.20.2.25 jump

- `/jump`

6.20.2.26 compass

- `/compass`

6.20.2.27 glow

- `/glow`

6.20.2.28 freeze

- `/freeze`

6.21 command_warmup

6.21.1 Purpose

This module adds a cooldown before command execution.

Note: The difference between command warmup and command cooldown

command warmup is before command execution, while command cooldown is after that.

6.21.2 Example

Example: Set warmup for all commands

```
{
  "command": {
    "regex": ".+",
    "ms": 3000
  },
  "interruptible": {
    "enable": true,
    "interrupt_distance": 3.0,
    "interrupt_on_damaged": true,
    "interrupt_in_combat": true
  }
}
```

Example: Set warmup for all commands except the `/back` command with a special setup

Since the rules are matched from up to down in order, you can just put the special case above.

```
[
  {
    "command": {
      "regex": "back",
      "ms": 10000
    },
    "interruptible": {
      "enable": true,
      "interrupt_distance": 3.0,
      "interrupt_on_damaged": true,
      "interrupt_in_combat": true
    }
  },
  {
    "command": {
      "regex": ".+",
      "ms": 3000
    },
    "interruptible": {
      "enable": true,
      "interrupt_distance": 3.0,
      "interrupt_on_damaged": true,
      "interrupt_in_combat": true
    }
  }
]
```

Example: Set warmup for all commands except the `/back` command

Use `negative lookahead` to exclude a string that starts with `"back"`.

```
{
  "command": {
    "regex": "(?!back).+",
    "ms": 3000
  },
  "interruptible": {
    "enable": true,
    "interrupt_distance": 3.0,
    "interrupt_on_damaged": true,
    "interrupt_in_combat": true
  }
}
```

Example: Share the same setup for multiple commands

Use the `or` clause to share the same setup.

```
{
  "command": {
    "regex": "(back)|(heal)|(feed)",
    "ms": 3000
  },
  "interruptible": {
    "enable": true,
    "interrupt_distance": 3.0,
    "interrupt_on_damaged": true,
    "interrupt_in_combat": true
  }
},
,
```

6.22 deathlog

6.22.1 Purpose

This module logs the inventory on player death.

6.22.2 Command

- `/deathlog`

6.22.3 Example

Example: Query the logs for a player

```
/deathlog view <player>
```

Tip: The death log index number is clickable

You can click the number to quickly restore the logged inventory to your inventory.

Example: Restore a death log from a player for a player

```
/deathlog restore <player> 0 <player>
```


6.23 disabler

6.23.1 Purpose

This module provides **disablers** to disable checkers inside **vanilla minecraft**.

6.23.2 Sub-Module

6.23.2.1 chat_speed_disabler

Disable **Kicked** for spamming.

6.23.2.2 move_speed_disabler

Disable **player moved too quickly** and **vehicle moved too quickly**.

6.23.2.3 move_wronlgy_disabler

Disable **player moved wrongly** and **vehicle moved wrongly**.

Warning: The movement anti-cheat inside vanilla minecraft is bad

Inside the vanilla minecraft server, there is a checker used to check if the player moves correctly. However, this checker usually makes wrong detection, and force setback the player, which makes the client-side gameplay feel lagged.

6.23.2.4 max_player_disabler

Disable the max players limit of the server.

6.24 echo

This module provides commands to send echo to players.

Note: Many other modules require this module

Other modules may generate a default configuration including the commands provided by echo module. If the echo module is disabled, then these echo commands will not exist, causing a command syntax error while executing these commands.

6.24.1 Sub-Module

6.24.1.1 send-message

- `/send-message`

Example: Say hello to a player

```
/send-message <player> Hello %player:name%
```

6.24.1.2 send-broadcast

- `/send-broadcast`

Example: Say hello to all players

```
/send-broadcast Hello %player:name%
```

6.24.1.3 send-actionbar

- `/send-actionbar`

Example: Say hello to a player

```
/send-actionbar <player> Hello %player:name%
```

6.24.1.4 send-title

- `/send-title`

Example: Send title to a player

```
/send-title <player> --mainTitle "<rainbow>Hello" --subTitle "<blue>World"  
--fadeInTicks 60 --stayTicks 60 --fadeOutTicks 60
```

Example: Send title to online players

```
/foreach send-title %player:name% --mainTitle "<rainbow>Hello %player:name%"
```

6.24.1.5 send-toast

- /send-toast

Example: Send toast to a player

```
/send-toast <player> --icon minecraft:golden_carrot <rb>eat this carrot
```

6.24.1.6 send-chat

- /send-chat

Example: Send chat as a player

```
/send-chat Steve i am steve.
```

Example: Send chat as a player for online players

```
/foreach send-chat %player:name% i am %player:name%
```

6.24.1.7 send-bossbar

- /send-bossbar

Example: A simple exapmle

```
/send-bossbar <player> Hello World
```

Example: All in one exapmle

```
/send-bossbar <player> --stepType BACKWARD --totalMs 5000 --color PURPLE
--style NOTCHED_6 --notifyMeOnComplete true --commandList "say the player
%player:name% is healed|heal others %player:name%" <rb>Healing is coming
[elapsed_time]/[total_time]/[left_time]
```

6.24.1.8 send-custom

- /send-custom

Example: Create a custom text

Create a plain text file named `"guide"` in `config/fuji/modules/echo/send_custom/custom-text/guide` with content:

```
<blue>===== custom text =====
Hello <orange>%player:name%</orange>, you are in
↳ <orange>%world:name%</orange> now.

<hover:show_text:'you see me!>Hover me</hover>

<click:run_command:'/back'>click me to run `/back` command</click>

<u><i><click:change_page:'3'>click me to the third page (this only works
↳ inside a book)</click></i></u>

<newpage><blue>This is the second page!

<click:suggest_command:'/back'>click me to suggest /back command (This
↳ doesn't work inside a book)</click>

<insert:'hello'>shift + click me to insert "hello" (This doesn't work
↳ inside a book)</insert>

<click:open_url:'https://placeholders.pb4.eu/user/text-format/'>click me
↳ to open the url</click>

<newpage>This is the third page!

<bold><click:change_page:'1'>click me to the first page</click></bold>

<orange>You can press `<keybind:'key.jump'>` key to jump!</orange>

<gradient:red:green:blue>This is gradient text.</gradient>

<rb>The rainbow text</rb>

<newpage>The end.
```

Example: Send custom text as a book

```
/send-custom as-book <player> guide --author "alice" --title "<rb>The Guide"
--giveBook true --openBook true
```

Example: Send custom text as a message

```
/send-custom as-message <player> guide
```

6.24.1.9 send-console

- `/send-console`

Example: Send a string to console

```
/send-console Hello World
```

6.25 fuji

6.25.1 Purpose

This module provides the command `/fuji`, which includes some operations on fuji itself.

6.25.2 Command

- `/fuji reload`
Reload all configuration files and all modules.

Note: Module itself can't be hot reloaded

After you enable or disable a module, you must restart the server.

- `/fuji about`
Open a gui to display the about, including the mod version and contributor list.
- `/fuji inspect modules`
Inspect all the enabled/disabled modules.
- `/fuji inspect server-commands`
Inspect all the registered commands in the server.
- `/fuji inspect fuji-commands`
Inspect all the commands registered by fuji mod.

Note: This will not show the requirement override from command permission module

The required level permission and required string permission are the default value set by fuji. If you are using `command_permission` module, then this gui will not show the overridden requirement of a command.

- `/fuji inspect argument-types`
Inspect all the registered argument types.
- `/fuji inspect configurations`
Inspect all the loaded configuration files.
- `/fuji inspect registry`
Inspect the registries in server.

6.26 functional

6.26.1 Purpose

This module allows players to open a virtual gui of functional-block.

6.26.2 Sub-Module

6.26.2.1 workbench

- `/workbench`

6.26.2.2 enchantment

- `/enchantment`

6.26.2.3 grindstone

- `/grindstone`

6.26.2.4 stonecutter

- `/stonecutter`

6.26.2.5 anvil

- `/anvil`

6.26.2.6 cartography

- `/cartography`

6.26.2.7 enderchest

- `/enderchest`

6.26.2.8 smithing

- `/smithing`

6.26.2.9 loom

- `/loom`

6.27 gameplay

6.27.1 Sub-Module

6.27.1.1 multi_obsidian_platform

6.27.1.1.1 Purpose

This module makes every **ender portal frame** generates its own **obsidian platform** (Up to 128 in survival-mode).

You can even use creative-mode to build more ender portal frame and more obsidian platform.

Note: All the obsidian platforms are vanilla-respect

All the extra obsidian platforms have the same behaviour as the vanilla one which locates in (100,50,0).

6.27.1.1.2 Configuration

factor The coordination-conversion factor between overworld and the_end. In vanilla minecraft, the factor between overworld and the_nether is 8.

6.27.1.2 carpet**6.27.1.2.1 Sub-Module****6.27.1.2.1.1 fake_player_manager**

Purpose Enable this module requires carpet-fabric mod installed. This module provides some management for fake-player.

Command

- `/player who`
- `/player renew`

Configuration

caps_limit_rule How many fake-player can each player spawn (in different time)?

The tuple means (day_of_week, minutes_of_the_day, max_fake_player_per_player).

The range of day_of_week is [1,7].

The range of minutes_of_the_day is [0, 1440].

For example: (1, 0, 2) means if the days_of_week ≥ 1 , and minutes_of_the_day ≥ 0 , then the max_fake_player_per_player now is 2.

Besides, you can add multi rules, the rules are checked from up to down.

The first rule that matches current time will be used to decide the max_fake_player_per_player.

You can issue `/player who` to see the owner of the fake-player.

Only the owner can operates the fake-player. (Op can bypass this limit)

renew_duration_ms How long should we renew when a player issue `/player renew` The command `/player renew` allows the player to manually renew all of his fake-player. If a fake-player don't gets renew, then it will expired and get killed. Use-case: to avoid some long-term alive fake-player.

transform_name The rule to transform the name of fake-player. Use-case: add prefix or suffix for fake-player.

6.27.1.2.1.2 better_info

Purpose Add nbt query for `/info` block command. Add the command `/info` entity.

6.28 head

6.28.1 Purpose

This module allows players to buy decorative heads from a head-database.

6.28.2 Command

- `/head`

6.28.3 Configuration

economy_type Can be ITEM or FREE

cost_type This option is used when economy type is ITEM, to specify which item as the currency.

cost_amount This option is used when economy type is ITEM, to specify the amount of currency to buy a head.

6.29 home

6.29.1 Purpose

This module allows players to set a teleportation point as their home.

6.29.2 Command

- `/home`

6.29.3 Meta

1. `fuji.home.home_limit` The home number per player limit.

6.30 kit

6.30.1 Purpose

This module allows you to make kits.

6.30.2 Concept

A **kit** is a set of itemstack.

6.30.3 Command

- `/kit editor`
Open the kit editor gui.
- `/kit give`
Give a kit to a player.

6.30.4 Example

Example: Create a kit

Use `/kit editor` to create a kit.

Note: The item slot position will keep

The item put inside the kit will keep its original position, so you can put armors in the right position.

Example: Give a kit to a player

`/kit give <player> <kit-name>`

Note: The giving function behaviour

1. try to insert the item in the specified slot
2. try to insert the item in any slot
3. drop the item in the ground with the player as its thrower

Example: Associate a cooldown to a kit

To associate a cooldown with commands: see [command_cooldown](#)

Create a named cooldown with 60sec cooldown and infinite usage: `/command-cooldown create example-kit-cooldown 60000`

Test the named cooldown, giving the kit to the player if the test is success: `/command-cooldown test example-kit-cooldown <player> --onFailed "send-message %player:name% wait a moment" kit give %player:name% example-kit|send-message %player:name% kit received.`

Now, create a new command using [command_bundle](#) module to execute the test command

Example: The config to create a new command for cooldown test command to give a kit if the test is success

```
{
  "requirement": {
    "level": 0,
    "string": null
  },
  "pattern": "claim-example-kit",
  "bundle": [
    "command-cooldown test example-kit-cooldown %player:name%
    ↪ --onFailed 'send-message %player:name% wait a moment
    ↪ (%fuji:command_cooldown_left_time example-kit-cooldown%
    ↪ ms)' kit give %player:name% example-kit|send-message
    ↪ %player:name% kit received."
  ]
}
```

Example: Claim a specific kit automatically for online players

You can use [command_scheduler](#) module to execute the `/claim-example-kit` for online players automatically every minute.

Example: Give a kit to newbie player automatically

See [command_event](#)

6.31 language

6.31.1 Purpose

This module provides client-side multi-language support.

Note: What is client-side language?

When the client joins a server, it will send its client options, including the client-side language value. The server can send messages in language used by the client later.

6.31.2 Feature

1. **Client-Side Respect** The client-side language will be respected if possible.
2. **Lazy-load** Only load the necessary language into the memory.

6.31.3 Difference

Disabled: All the players use the `default_language`.

Enabled: Fuji will **try** to respect the player's client-side language, if the server-side supports.

6.31.4 Example

Example: To suppress the sending of a message

Set the value of a message key to "`[suppress-sending]`", to suppress the sending of this type of message.

6.32 motd

6.32.1 Purpose

This module provides motd customization.

6.32.2 Example

Example: Configure server icons

You can put 64x64 pixels `.PNG` `images` into the directory `config/fuji/modules/motd/icon`, this module will pick up a random image as the icon of the server to respond the server status request.

Example: Configure random motd text

You can write multiple `motd text` in the configuration, the `motd text` will be chosen randomly.

6.32.3 Reference

1. <https://colorize.fun/en/minecraft>

6.33 multiplier

6.33.1 Purpose

This module provides some **numeric multiplier**.

6.33.2 Supported Numeric Types

1. **damage** damage to player
2. **experience** experience a player gained

6.33.3 Example

Example: Double the damage from zombie to a player

```
/lp group default meta set fuji.multiplier.damage.minecraft:zombie 2
```

Example: Cancel the fall damage

```
/lp group default meta set fuji.multiplier.damage.minecraft:fall 0
```

Example: Double all damage to a player

```
/lp group default meta set fuji.multiplier.damage.all 2
```

Example: Double all experience a player gained

```
/lp group default meta set fuji.multiplier.experience.all 2
```

Example: Half all damage to a player

```
/lp group default meta set fuji.multiplier.damage.all 0.5
```

6.34 nametag

6.34.1 Purpose

This module provides nametag customization.

6.34.2 Configuration

update_cron The cron used for the job to **update** the properties of **display entity**.

style Define the style for the nametag **display entity**.

render Define the nametag render behaviour.

Note: The explanation of each field in style and render

You can refer to the minecraft wiki about **display entity**: <https://minecraft.wiki/w/Display>

6.34.3 Example

Example: Set background to blue color

```
"background": -16776961
```

Example: Set half transparency

```
"text_opacity": 128
```

Example: Scale the size of text into double

```
"scale": {  
  "x": 2.0,  
  "y": 2.0,  
  "z": 2.0  
},
```

6.35 placeholder

6.35.1 Purpose

This module provides more `placeholder` for `Text Placeholder API` mod.

6.35.2 Command

- `/placeholder`

6.35.3 Placeholder

1. `%fuji:player_mined%` sum of mined blocks of a player
2. `%fuji:server_mined%` sum of mined blocks of a server
3. `%fuji:player_placed%` sum of placed blocks of a player
4. `%fuji:server_placed%` sum of placed blocks of a server
5. `%fuji:player_killed%` sum of killed entities of a player
6. `%fuji:server_killed%` sum of killed entities of a server
7. `%fuji:player_moved%` sum of moved distance of a player
8. `%fuji:server_moved%` sum of moved distance of a server
9. `%fuji:player_playtime%` sum of playtime of a player
10. `%fuji:server_playtime%` sum of playtime of a server
11. `%fuji:health_bar%` the health bar of a player
12. `%fuji:rotate hello%` rotate the string `hello`
13. `%fuji:has_permission%` check luckperms permission
14. `%fuji:has_meta%` get luckperms meta
15. `%fuji:random_player%` get a random online player
16. `%fuji:random 1 5%` get a random number from 1 to 5
17. `%fuji:escape%` escape a placeholder from the parser. An optional number argument is used as the levels to escape.
18. `%fuji:protect%` protect a string from the parser.
19. `%fuji:date%` get current date.
An optional string argument is used to set the **date formatter**, for example, `%fuji:date HH:MM%`.
See also: <https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html>
20. `%fuji:player_prefix%` player luckperms prefix
21. `%fuji:player_suffix%` player luckperms suffix
22. `%fuji:pos%` player current position

6.35.4 What's more?

Tip: Use placeholder in language file

It's allowed to write placeholders in language file.

Note: Some other mods that provide more placeholders

<https://placeholders.pb4.eu/user/mod-placeholders/>

6.36 predicate

6.36.1 Purpose

This module provides `predicate` commands.

6.36.2 Definition

A command that suffixes with the character `"?"` is named `predicate` command.
e.g. The `/has-perm?` command.

6.36.3 How it works?

The `return` value is represented in an `integer` provided by vanilla minecraft command return layer.
This command will return 1 if test is success, and -1 if test is failed.

See more: <https://minecraft.fandom.com/wiki/Commands/return>

6.36.4 Example

Example: Test a condition and then run a command

See more in [chain](#)

```
/run as player <player> chain has-perm? %player:name% 4 chain say value is true
```

6.37 profiler

6.37.1 Purpose

This module shows server health status, such as os, vm, cpu, ram, tps, mspt and gc.

Warning:

Enable this module requires spark mod installed.

6.37.2 Command

- `/profiler`

6.38 pvp

6.38.1 Purpose

This module provide pvp state toggle.

6.38.2 Command

- `/pvp`

6.39 rtp

6.39.1 Purpose

Provides random teleportation.

6.39.2 Feature

1. Per dimension configurable.
2. Ignore fluid blocks, such as water and lava.
3. Ignore powered snow.

6.39.3 Command

- `/rtp`

6.39.4 Configuration

setup Teleport setup per dimension. Dimensions that are not in the list will be disabled for rtp.

6.39.5 What's more?

Tip: Improve the performance on rtp

It's highly recommended to pre-gen the world chunks. To gen a new chunk during rtp requires about 2 to 10 seconds. If a chunk is pre-gen, then it will be fast.

6.40 skin

6.40.1 Purpose

This module provides skin customization.

6.40.2 Command

- `/skin`

6.40.3 Configuration

default_skins The default skin used for player who has no skin set.

6.40.4 Example

Example: Set a skin from mojang

```
/skin set mojang dream
```

Example: Set a skin from url

```
/skin set web slim "https://s.namemc.com/i/bd53d152d0cd91d0.png"
```

Example: Use default skins

```
/skin use-default-skins
```

Example: Use online skin

```
/skin use-online-skin
```

6.41 system_message

6.41.1 Purpose

This module provides **system message** customization.

6.41.2 How it works?

The server will send **translatable text** to the client, by default, the client will display the translatable text in the language used in client-side. This module will hijack the pre-define **translatable text** in the server-side, and sends the client a **customized text** directly. As a loss of this approach, the client will always receive the customized text defined in the server, instead of the **translatable text** defined in the client.

Note: What is translatable text?

For example:

1. player join and leave server message
2. player advancement message
3. player death message
4. player command feedback
5. player whitelist message

Note: How to query the translatable text?

See [mojang's official language file](#), all keys inside the file are for **translatable text**. Note that, some keys in the file are client-side only translatable text, which means that the server will never send this type of translatable text to the client.

Warning: Possible conflicting with StyledChat

If you are using this module with **StyledChat** mod installed, you need to remove the conflicting **language keys** in **key2value** configuration.

See also: <https://github.com/sakurawald/fuji/issues/65>

See also: <https://github.com/sakurawald/fuji/issues/203>

6.41.3 Example

Example: Custom the player join and leave message.

```
"system_message": {
  "enable": true,
  "key2value": {
    "multiplayer.player.joined": "<rainbow>+ %s",
    "multiplayer.player.left": "<dark_gray>%s leeeeeeeft the game"
  }
},
```

Example: Custom a specific death message.

```
"system_message": {
  "enable": true,
  "key2value": {
    "death.attack.explosion.player": "<rainbow>%1$s boooooooooom because
    ↪ of %2$s"
  }
},
```

Example: Custom messages used in screen.

```
"system_message": {
  "enable": true,
  "key2value": {
    "multiplayer.disconnect.server_shutdown": "Server closeeeeeeeeed",
    "multiplayer.disconnect.not_whitelisted": "<rainbow>Please apply a
    ↪ whitelist first!"
  }
},
```

Example: Custom messages in a container

```
"key2value": {
  "container.chest": "<rb>I see you opening the chest!"
}
```

Example: Custom the vanilla command feedback.

```
"system_message": {  
  "enable": true,  
  "key2value": {  
    "commands.seed.success": "<rainbow> Seeeeeeeeeeed: %s"  
  }  
},
```

Example: Custom the player disconnect message

```
"system_message": {  
  "enable": true,  
  "key2value": {  
    "multiplayer.player.left": "<dark_gray>%s leeeeeeeft the game"  
  }  
},
```

Example: Cancel the player disconnect message

```
"system_message": {  
  "enable": true,  
  "key2value": {  
    "multiplayer.player.left": null  
  }  
},
```

6.42 tab

6.42.1 Purpose

This module provides tab list customization.

6.42.2 Configuration

update_cron The cron used for the job to update the tab list.

style The style for tab list.

Note: How to design a complex tab list?

For complex tab list customization, I would recommend to use <https://github.com/NEZNAMY/TAB>

6.43 teleport_warmup

6.43.1 Purpose

This module adds a warmup cooldown before player-teleportation.

6.43.2 Configuration

warmup_second The second to wait before the teleportation.

interruptible Should we interrupt this teleporation if some conditions meet?

dimension Per dimension configuration.

blacklist Only apply teleport warmup in the following dimensions.

Warning: Dimensions that created by other mods may have special behaviour

Some other mods will add extra dimension (like, the mod the-bumblezone-fabric). Their dimension portal may work in a different way, so this module may not be compatible with these mods.
In the default options, we only allow this module works in the vanilla minecraft dimensions.

6.43.3 Meta

1. fuji.teleport_warmup.warmup The per player teleport warmup time in seconds.

6.44 temp_ban

6.44.1 Purpose

This module provides temp-ban.

6.44.2 Command

- `/temp-ban`

6.44.3 Example

Example:

```
/temp-ban player <player> 1s2m3h4d5w6M7y bad boy
```

6.45 tester

6.45.1 Purpose

This module is only used for development. If you are a developer, you can register new commands into this module for test-purpose.

6.45.2 Command

- `/tester`

Warning:

You should not use this module at the production-environment, because it almost does nothing useful.

6.46 top_chunks

6.46.1 Purpose

This module compute a **laggy score** for all loaded chunks, and return the topN lagged chunks. Higher score means more lagged.

6.46.2 Command

- `/chunks`

6.46.3 Configuration

top The top chunks to show in `/chunks` command

nearest_distance For a chunk, how much the radius used to search the nearest player around the chunk.

hide_location Should we hide the chunk-position for a lagged-chunk? Hide chunk location to avoid grief or privacy purpose.

type2score The dict to define how lagged a type(entity/entity_block) should be.
For example:

Example: What is the meaning of type2score field?

The configuration means that if there are 15 zombies inside a chunk, then the chunk gets score $15 * 4 = 60$. Any other types not specified in type2score will use the score defined for type **default**.

```
"type2score": {  
  "entity.minecraft.zombie": 4,  
  "default": 1  
}
```

6.47 tpa

6.47.1 Purpose

This module provides teleport request for players.

6.47.2 Command

- `/tpa`
- `/tpahere`
- `/tpaaccept`
- `/tpadeny`
- `/tpacancel`

6.47.3 Configuration

timeout Tpa request expiration duration, unit is second.

mention_player See [6.4.3.3.2](#)

6.48 view

6.48.1 Purpose

This module provides the player slot editor.

6.48.2 Command

- `/view inv`
- `/view ender`

6.49 whitelist

6.49.1 Purpose

This module makes the mojang vanilla whitelist system only compares the **username** and **ignore UUID**.

Warning: Only enable this module in offline-mode server

If you are hosting a online-mode server, you will never need to enable this module.

6.50 works

6.50.1 Purpose

This module provides a bill-board gui for players to show their **works**.

6.50.2 Concept

A **work** is a **teleporation point**.

The types of work are as follows:

1. **Non-Production-Work** the project don't produce any resource (e.g. bone, string, coal).
2. **Production-Work** the project produce some resource.

Note: The main difference between non-production work and production-work

For a production-work, fuji provides the **production sample** to count the **hopper** and **minecart-hopper**

Tip: About the production counter

1. You can use the production counter provided by production work to sample the output.
2. This module works with carpet-fabric's hopper counter. You can use both of them at the same time.
3. The hopper counter provided by this module will not destroy the item.

6.50.3 Command

- `/works`

6.50.4 Configuration

sample_time_ms For a production-work, how long should we sample it?

sample_distance_limit For a production-work, how large the radius should we considered as the work's production.

sample_counter_top_n For a production-work, we only display the topN output items.

6.51 world

6.51.1 Purpose

Provides a unified world management.

6.51.2 Command

- `/world`

6.51.3 Concept

Note: What is the difference between world, dimension and dimension type?

Well, in the early stage of minecraft, a **world** only support **single-dimension**, which means 1 world only contains 1 dimension.

But now, 1 world supports multi dimension. Sometimes, you will see **world** and **dimension** means the same thing.

But clearer, we say: 1 world can contain 1 or more dimension, and each dimension has its **dimension type**.

Usually, you can say a mod adds extra dimension type and **creates an extra dimension** with that dimension type instead of **creating extra world**.

See also: https://minecraft.wiki/w/Dimension_definition

See also: https://minecraft.wiki/w/Dimension_type

Note: The dimension and dimension types in vanilla minecraft

In vanilla minecraft, 1 world contains 3 dimensions:

1. minecraft:overworld
2. minecraft:the_nether
3. minecraft:the_end

You can see the dimensions of a world in **world/level.dat** file.

A dimension type is used to create dimensions, the vanilla minecraft has the following dimension type:

1. minecraft:overworld
2. minecraft:overworld_caves
3. minecraft:the_nether
4. minecraft:the_end

The file **server.properties** is used for **the only and default world**.

6.51.4 Configuration

blacklist The dimensions in the blacklist will not be operated by this module. Use blacklist to avoid mis-operation.

6.51.5 Example

Example: Create an extra the_nether dimension

```
/world create my_nether minecraft:the_nether
```

Example: Delete the extra dimension

```
/world delete fuji:my_nether
```

Example: Reset the extra dimension with random seed

```
/world reset fuji:my_nether
```

Example: Specify a seed for an extra dimension.

```
/world create my_nether --seed 1234567890 minecraft:the_nether  
/world reset --useTheSameSeed true fuji:my_nether
```

Tip: Make a resource-world that automatically reset every day

You can use `command_scheduler` module to execute `/world reset` command automatically.

6.52 world_downloader

6.52.1 Purpose

This module allows a player to download nearby chunks.

6.52.2 Command

- `/download`

6.52.3 Configuration

url_format The url format used to broadcast.

port The port used for downloader http-service.

bytes_per_second_limit Max download speed for each connection.

context_cache_size Max download request saved in the memory.

Chapter 7

Q&A

7.1 Where is the configuration files?

As a convention, all the files are placed in `config/fuji/` directory.

7.2 What is .json file?

A json file is a text file, whose name normally ends with `.json`.

7.3 How can I edit a configuration file?

To ensure the readable and transparent, most of the files are saved as pure text format. You can open them with a text editor.

Tip: Use a modern text editor.

Some files may have a large number of lines, so it's highly recommended to use a modern text editor, which can highlight symbols and reveal the structure of the file, such as:

1. Visual Studio Code
2. Visual Studio Code - Web Online Editor
3. Vim
4. Emacs
5. Sublime Text

7.4 What is .dat file?

The file whose name ends with `.dat` are the vanilla minecraft NBT format file. To open such a file, you need to use a NBT Editor, such as NBTEditor.

7.5 How to update fuji to a new version?

7.5.1 Backup the data

Back up the config/fuji directory.

7.5.2 Test the new version in your test-environment

Put the new version of fuji into mods/ directory, start the server, and adjust the configuration to what you want.

Warning: Don't test new changes directly in your production-environment

It's highly recommended to setup a test-environment for a network-maintainer, so that you can test and tweak installed mods into what you want, and avoid un-expected situations.

7.5.3 Apply the changes to production-environment

Now, it's ready to apply the changes to your production-environment.

7.6 Fuji conflits with one of my mods.

You can disable the conflicting module. If possible, create an issue at [fuji issue page](#), so that we can solve this later.

7.7 Can I ask the forge or neoforge support?

We have no plan for forge or neoforge platform. However, you can try running this mod via **sinytra-connector** mod. It's tested the mod works in the following environment (105/105 modules works):

```
Loader: NeoForge v21.1.73:server for Minecraft 1.21.1
Mods:
- connector-2.0.0-beta.3+1.21.1-full.jar
- forgified-fabric-api-0.104.0+2.0.15+1.21.1.jar
- fuji-4.3.0-5518b05201-mc1.21.jar
```

7.8 How can I report bugs or suggest new features?

You can create an issue at [fuji issue page](#)

Chapter 8

Transformer

8.1 Command Transformer

The following commands can be transformed:

1. `/blockcycling` = `/give <player> minecraft:debug_stick`
2. `/blockinfo` = use the command `/info block` provided by `carpet` mod.
3. `/blocknbt` = `/data get block`
4. `/entityinfo` = `/data get entity`
5. `/entitynbt` = `/data get entity`
6. `/customrecipe` = `recipe editor`
7. `/exp` = `/experience`
8. `/gm`, `/gms`, `/gms` = `/gamemode`
9. `/findbiome` = `/locate biome`
10. `/tempfly` = use `luckperms temporal permission`
11. `/flyspeed` = `/attribute <player> minecraft:generic.flying_speed`
12. `/walkspeed` = `/attribute <player> minecraft:movement_speed`
13. `/maxhealth` = `/attribute <player> minecraft:generic.max_health`
14. `/groundclean` = `/kill @e[type=...]`
15. `/spawner` = interact `spawner block` with `spawn egg`
16. `/spawnmob` = `/summon`
17. `/shoot` = `/summon` with `motion`
18. `/smite`, `/thunder` = `/summon minecraft:lightning_bolt`

19. `/stats`, `/statsedit` = all stats files are located in `world/stats`
20. `/tree` = use `tree brush` from `world-edit` mod
21. `/unbreakable` = `/enchant <player> minecraft:unbreaking`
22. `/item_editor` = `/give` with command generator, or use some client-side mod to edit items with commands.
23. `/replaceblock` = `/setblock`
24. `chunks loading/unloading commands` = These commands will not bring performance improvements.
25. `/note` = `/attachment get note`
26. `/...all` = `/foreach ...`
27. `/cuff` = use `anti-build` module with `string` permission
28. `/cheque` = use `command-attachment` module to make magic item

Tip: Use `command bundle` module to create transformed commands

See [command_bundle](#)

8.2 Command Generator

You can generate complex and powerful commands using:

1. <https://www.digminecraft.com/generators/>
2. <https://minecraft.tools/en/public-gallery.php>

Chapter 9

Development

9.1 Setup the development environment

1. Clone the source:

```
git clone https://github.com/sakurawald/fuji.git
```

2. Change the working-directory:

```
cd fuji
```

3. Compile the source:

```
./gradlew build
```

Chapter 10

Suggestion

10.1 Suggestion on server-side mods

10.1.1 Explanation

Here are some mods that existing and recommended to use, which can make your life easier in fabric server-side crafting.

Note that fuji doesn't require these mods installed to work, and some of these mods have the same functionality as fuji.

If you want to taste something different, and if some of the mods provides a better experience than fuji, then you can just disable the module in fuji, and go use it.

Tip: Decide according to your situation

The suggestion is personal, you should make decision according to your preference.

10.1.2 Server-side mode list

1. [blue map](#)
2. [anti xray](#)
3. [melius commands](#)
4. [vanilla permission](#)
5. [vanish](#)
6. [villager config](#)
7. [world-edit](#)
8. [essential commands](#)
9. [missions](#)
10. [luckperms](#)
11. [tab](#)
12. [armor stand editor](#)
13. [ban hammer](#)
14. [image2map](#)
15. [polydecorations](#)
16. [sleep warp](#)
17. [styled chat](#)
18. [styled nickname](#)
19. [styled player list](#)
20. [styled sidebar](#)
21. [universal graves](#)
22. [universal shop](#)
23. [goml](#)
24. [polydex](#)
25. [polymania](#)
26. [head index](#)
27. [inv view](#)
28. [ledger](#)

- 29. [falling tree](#)
- 30. [double doors](#)
- 31. [skin restorer](#)
- 32. [husk homes](#)
- 33. [yet another world protector](#)
- 34. [krypton](#)
- 35. [sit](#)
- 36. [stack deobf](#)
- 37. [lithium](#)
- 38. [let me despawn](#)
- 39. [carpet](#)
- 40. [mod viewer](#)
- 41. [simple voice chat](#)
- 42. [mini motd](#)
- 43. [tab tps](#)
- 44. [spark](#)
- 45. [polyfactory](#)
- 46. [ouch](#)
- 47. [chunky](#)
- 48. [afk plus](#)
- 49. [taterzens](#)